

Question number	Answer	Mark		
5(a)(i)	The only correct answer is C. <table><tr><td>fructose and glucose</td><td>sucrose only</td></tr></table> <i>A is incorrect because glucose is also a monosaccharide not a disaccharide</i> <i>B is incorrect because fructose is also a monosaccharide and not a disaccharide</i> <i>D is incorrect because sucrose is not a monosaccharide and fructose is not a disaccharide</i>	fructose and glucose	sucrose only	(1)
fructose and glucose	sucrose only			

Question number	Answer	Mark
5(a)(ii)	The only correct answer is B glycosidic	

	<i>A is incorrect because ester bonds join organic acids and alcohols together</i> <i>C is incorrect because peptide bonds join amino acids together</i> <i>D is incorrect because phosphodiester bonds joins mononucleotides together</i>	(1)
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Question number	Answer	Additional guidance	Mark
5(b)(i)	A calculation showing the following steps: - the mean calculated (1) - the mean converted to g dm^{-3} (1)	Example of calculation : $(1.22 + 1.02 + 1.00) \div 3 = 1.08$ $0.00108 / 0.0011 = 1.08 \times 10^{-3}$ Correct answer with no working shown gains both marks.	(1)

Question number	Answer
*5(b) (ii)	Indicative content: • sugar solution with a lower water potential than the cytoplasm of the cell (D) • sugar solution should be hypertonic (D) • so that water will pass out of the cytoplasm by osmosis (E) • concentration of sugar solution should be the same as the cytoplasm (D) • so that sugars will not diffuse out (E) • each individual sugar concentration should be the same as the cytoplasm (D) • mean concentration of sugars stated (fructose = 1.52, glucose = 1.08, sucrose = 8.55) (D) • range of sugar concentrations stated (from table) (D)

- so that concentration of each sugar remains the same (E)
- another solute needs to be used (D)
- so that the sugar concentration remains the same but the water passes out (E)

Level 1 : description of conditions needed for osmosis and maintaining sugar concentration

1 mark = a description of one condition

2 marks = a description of each condition or an explanation of one condition

Level 2 : explanation of conditions needed for both osmosis and maintaining sugar concentration

3 marks = a description of one condition and an explanation of the other condition

4 marks = an explanation of each condition

Level 3 : indicates that concentration of each sugar needs to be the same as in the pineapple and another solute needs to be included.

5 marks = and includes one of the above

6 marks = and includes both of the above